University of Nottingham
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Python基础三

2026

Learning outcomes

By the end of this session, you should be able to:

- 数据清洗—异常值和重复值
- 数据转换与数据重塑

处理异常值

如何处理异常值：

- **情景一：常规识别**
- **情景二：Zscore识别**

处理异常值

情景一：常规识别

处理异常值

情景1：

绝对值大于5为异常

识别异常值

outlier=df[df[np.abs(df)>5].any(1)]

删除异常值

df[~df.index.isin(outlier.index)]

	rating_A	rating_B
A001	-1	-5
A002	3	5
A003	4	4
A004	6	5
A005	3	5
A006	5	5

[32]

```
#识别异常值
outlier=df[df[np.abs(df)>5].any(1)]
```

[33]

```
print(outlier)
```

```
       rating_A  rating_B
A004          6         5
```

[34]

```
#删除异常值
df[~df.index.isin(outlier.index)]
```

	rating_A	rating_B
A001	-1	-5
A002	3	5
A003	4	4
A005	3	5
A006	5	5

处理异常值

情景二：Zscore识别

处理异常值

情景2：

使用Z-score判断异常

	rating_A	rating_B
A001	-1	-5
A002	3	5
A003	4	4
A004	6	5
A005	3	5
A006	5	5

```python
#情景2: 使用Z-score 判断异常
# zscore.abs>2--> outlier
#识别异常值

for col in df.columns:
    df_current_col=df[col]
    z_score=(df_current_col-df_current_col.mean())/df_current_col.std()
    df_zscore=z_score.abs()>2

outlier=df[df_zscore]
print(outlier)

#删除异常值
drop_outlier=df[df_zscore==False]
drop_outlier
```

处理重复值

识别重复值

df [df.duplicated()]

删除重复值

df.drop_duplicates(['score'])

```python
import pandas as pd
import numpy as np
df = pd.DataFrame({
    'studentID':['A001', 'A002', 'A003','A004', 'A005', 'A006', 'A006'],
    'score': [100, 93, 94,96 , 93, 95, 95] }
)
df
```

	studentID	score
0	A001	100
1	A002	93
2	A003	94
3	A004	96
4	A005	93
5	A006	95
6	A006	95

处理重复值

识别重复值

df [df.duplicated()]

删除重复值

df.drop_duplicates(['score'])

```
#如何删除
# df.drop_duplicates()
#A005同学的成绩因为和A002同学的相同所以被删除了
#利用子集进行去重
df.drop_duplicates(['score'])
```

	studentID	score
0	A001	100
1	A002	93
2	A003	94
3	A004	96
5	A006	95

```
df.drop_duplicates(keep=False,inplace=True)
```

```
df
```

	studentID	score
0	A001	100
1	A002	93
2	A003	94
3	A004	96
4	A005	93

Practice 1

```python
import pandas as pd
import numpy as np
df = pd.DataFrame({'rating_A': [-1, 3, 4,6, 3,5],
'rating_B': [-5, 5, 4,5, 5, 5]},
index=['A001', 'A002', 'A003','A004', 'A005', 'A006'])
df
```

	rating_A	rating_B
A001	-1	-5
A002	3	5
A003	4	4
A004	6	5
A005	3	5
A006	5	5

在该数列上进行异常值和重复值的识别与删除操作

DataFrame 切片获取

columns 获取DataFrame的列

print(df.columns)

index 获取DataFrame的索引

print(df.index)

```python
import pandas as pd
import numpy as np
df = pd.DataFrame({
'score_a': [80, 93, 94,96 , 90, 92, 93],
'score_b': [80, 93, 94,96 , 99, 92, 90],
'score_c': [99, 93, 94,96 , 93, 95, 92]}
,index= ['A001', 'A002', 'A003','A004', 'A005', 'A006', 'A007']
)
df
```

	score_a	score_b	score_c
A001	80	80	99
A002	93	93	93
A003	94	94	94
A004	96	96	96
A005	90	99	93
A006	92	92	95
A007	93	90	92

DataFrame 切片获取

转换为列表类型
print(list(df.index))

columns 获取DataFrame的列
print(list(df.columns))

index 获取DataFrame的索引
print(list(df.index))

取行
前两行
df[:2]
开始行到倒数第二行
df[:-1]
取列
df['score_a']
行&列并取
#取 前两行 的 'score_a'列
df[:2]['score_a']

DataFrame 切片获取

loc / iloc

- loc：works on labels in the index.作用在索引上
- iloc：works on the positions in the index (so it only takes integers).作用在索引的位置上

```
#取行：A001行
#loc
df.loc['A001']
```

```
score_a    80
score_b    80
score_c    99
Name: A001, dtype: int64
```

```
#A001
#iloc
df.iloc[0,:]
```

```
score_a    80
score_b    80
score_c    99
Name: A001, dtype: int64
```

DataFrame 切片获取

- 取不连续多行：
- 取A001和A003这两位同学的成绩
- loc
- df.loc[['A001','A003'],:]

- 取不连续多行：
- 取A001和A003这两位同学的成绩
- iloc
- df.iloc[[0,2],:]

DataFrame 切片获取

根据条件获取：
例子：取a成绩大于90
方法1:

```
df[df['score_a']> 90]['score_a']
```

方法2:

```
df.loc[df['score_a']> 90,'score_a’]
```

方法3:

```
df.iloc[(df['score_a']> 90).values,0]
```

Thanks for listening!